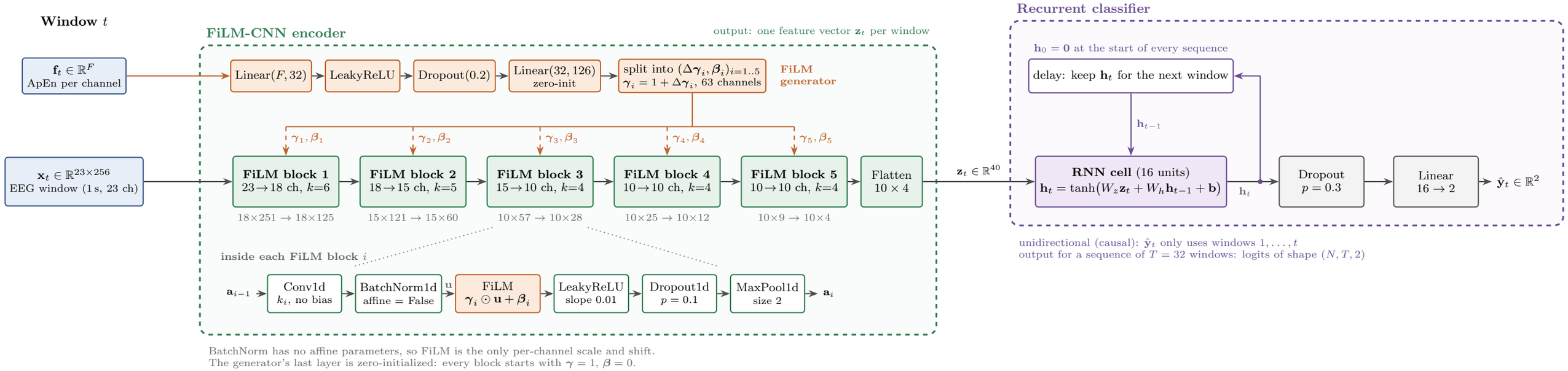

Window t
$\mathbf{f}_t \in \mathbb{R}^F$
ApEn per channel
$\mathbf{x}_t \in \mathbb{R}^{23\times256}$
EEG window (1 s, 23 ch)
FiLM-CNN encoder
output: one feature vector $\mathbf{z}_t$ per window
Linear$(F, 32)$
LeakyReLU
Dropout(0.2)
Linear(32, 126)
zero-init
split into $(\Delta\boldsymbol{\gamma}_i, \boldsymbol{\beta}_i)_{i=1..5}$
$\boldsymbol{\gamma}_i = 1 + \Delta\boldsymbol{\gamma}_i$, 63 channels
FiLM generator
$\boldsymbol{\gamma}_1, \boldsymbol{\beta}_1$
$\boldsymbol{\gamma}_2, \boldsymbol{\beta}_2$
$\boldsymbol{\gamma}_3, \boldsymbol{\beta}_3$
$\boldsymbol{\gamma}_4, \boldsymbol{\beta}_4$
$\boldsymbol{\gamma}_5, \boldsymbol{\beta}_5$
FiLM block 1
23→18 ch, k=6
18×251 → 18×125
FiLM block 2
18→15 ch, k=5
15×121 → 15×60
FiLM block 3
15→10 ch, k=4
10×57 → 10×28
FiLM block 4
10→10 ch, k=4
10×25 → 10×12
FiLM block 5
10→10 ch, k=4
10×9 → 10×4
Flatten
10 × 4
$\mathbf{z}_t \in \mathbb{R}^{40}$
inside each FiLM block i
$\mathbf{a}_{i-1}$
Conv1d
k_i, no bias
BatchNorm1d
affine = False
u
FiLM
$\boldsymbol{\gamma}_i \odot \mathbf{u} + \boldsymbol{\beta}_i$
LeakyReLU
slope 0.01
Dropout1d
$p = 0.1$
MaxPool1d
size 2
$\mathbf{a}_i$
BatchNorm has no affine parameters, so FiLM is the only per-channel scale and shift.
The generator's last layer is zero-initialized: every block starts with $\boldsymbol{\gamma} = 1$, $\boldsymbol{\beta} = 0$.
Recurrent classifier
$\mathbf{h}_0 = \mathbf{0}$ at the start of every sequence
delay: keep $\mathbf{h}_t$ for the next window
$\mathbf{h}_{t-1}$
RNN cell (16 units)
$\mathbf{h}_t = \tanh(W_z\mathbf{z}_t + W_h\mathbf{h}_{t-1} + \mathbf{b})$
$\mathbf{h}_t$
Dropout
$p = 0.3$
Linear
16 → 2
$\hat{\mathbf{y}}_t \in \mathbb{R}^2$
unidirectional (causal): $\hat{\mathbf{y}}_t$ only uses windows $1, \ldots, t$
output for a sequence of $T = 32$ windows: logits of shape $(N, T, 2)$